



Northeastern University
College of Professional Studies

EAI6020 - AI Systems Technology

Module 4 Assignment

Leveraging AutoML for Credit Card Fraud Detection

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Introduction

This report is using Google Cloud's Vertex AI AutoML platform. The goal was to tackle a classic and critical business problem: credit card fraud detection. This project provided a chance to move beyond theoretical concepts and engage directly with the challenges and decision-making steps involved in deploying a machine learning model. The process involved selecting a public dataset, setting up a clear business case with economic variables, and navigating the real-world operational challenges of a cloud AI platform.

About Dataset

For this assignment, I chose the [Credit Card Fraud Dataset](#) from Kaggle. It represents a business problem, as [\[1\] Chaudhary. all \(2012\)](#) note that credit card fraud is a major issue for financial institutions. Most importantly, the dataset has a severe class imbalance, with only 1% of transactions being fraudulent. This makes simple accuracy a poor metric and highlights the need for the precision-recall curve.

Here's a simple business scenario to define the economic viability of the AI solution:

- **Cost of a False Negative (Missing a Fraudulent Transaction):** \$500. This is the direct financial loss.
- **Cost of a False Positive (Incorrectly Flagging a Legitimate Transaction):** \$50. This shows the customer support costs and brand damage from bothering a valid customer.

This clear economic trade-off means the model's success must be measured by its ability to minimize these costs, not just by its technical accuracy.

How I Trained and Evaluated the Model using the Precision-Recall Curve

The process of training the model on Vertex AI provided a valuable lesson in the realities of using cloud-based MLOps.

The Training Process Record: From Failure to Success

My first attempt to train the model used the full dataset of 100,000 transactions. This attempt **failed** with a QUOTA_FOR_INSTANCES error. This meant that the resources needed by Vertex AI's background processes exceeded my project's default limits. This experience was a practical example of the challenges discussed by [\[3\] Wang. all \(2024\)](#), who point out that managing the cost and resource utilization of platforms like Vertex AI is a critical competency.

To fix this, I used stratified sampling, a common data engineering solution. I created a smaller sample of 20,000 rows, making sure to keep the original 1% fraud ratio. This smaller dataset was then used for a new training job.

The second training attempt was set up as follows:

- **Dataset:** fraud-detection-sample-20k
- **Target Column:** IsFraud
- **Optimization Objective:** AUC PR. I chose this to make AutoML focus on the model's performance on the rare fraud cases.
- **Budget:** 1 node hour.

This training job was successfully completed after a total training time of 2 hours and 15 minutes. Here's some screenshots for evaluation details:

Evaluation details

Confidence threshold 

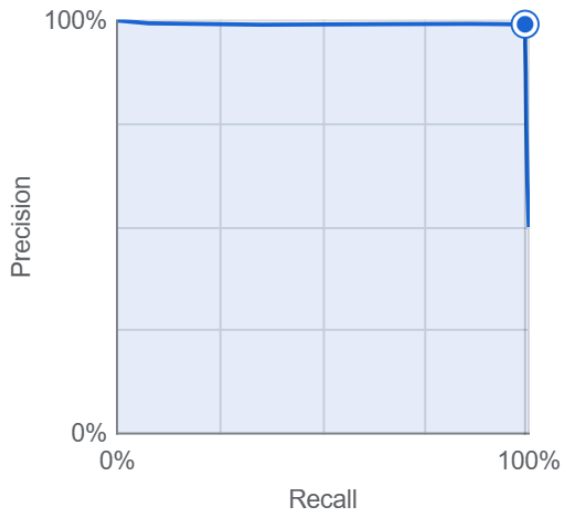


0.5

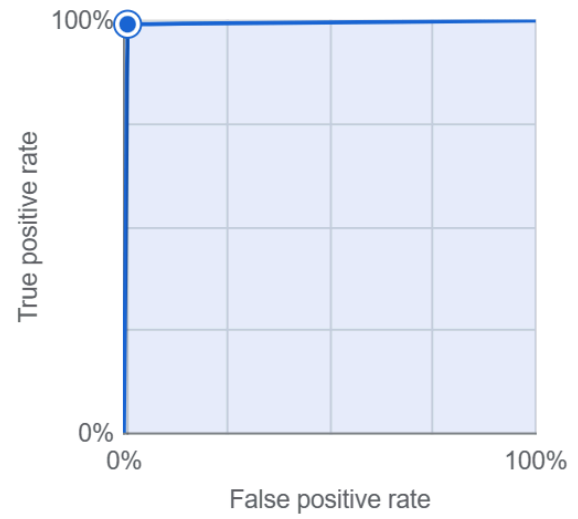
All labels

PR AUC 	0.989
ROC AUC 	0.991
Log loss 	0.192
Micro-average F1 	0.99114174
Macro-average F1 	0.49777558
Micro-average precision 	99.1%
Micro-average recall 	99.1%
Total items	0
Training items	0
Validation items	0
Test items	0

Precision-recall curve ?



ROC curve ?



Confusion matrix

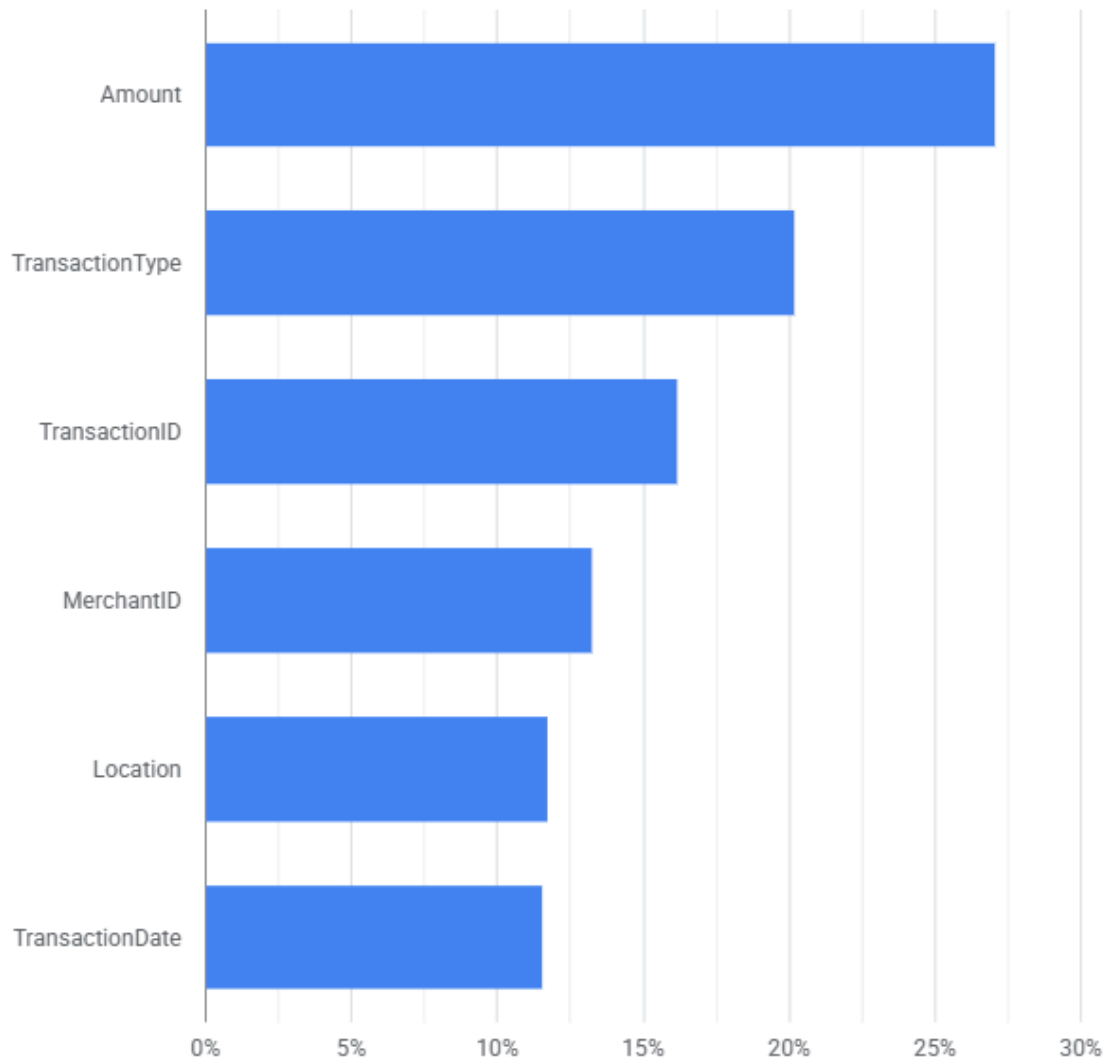
☒ Item counts

A confusion matrix shows how the model classified each label in the evaluation dataset. The blue, bold cells indicate a correct prediction. A data item is moved to the dropped column if it does not meet the confidence threshold for any label.

True label	Predicted label		
	0	1	Dropped
0	100%	0%	0%
1	100%	0%	0%

Feature importance

Model feature attribution tells you how important each feature is when making a prediction. Attribution values are expressed as a percentage; the higher the percentage, the more strongly that feature impacts a prediction on average. Model feature attribution is expressed using the Sampled Shapley method. [Learn more](#) 



Model Evaluation

The final model was very high-quality, achieving a **PR AUC of 0.989** and a **ROC AUC of 0.991**.

However, at the default 0.5 confidence threshold, the **Macro-average F1 score was only 0.498**. This low score was a clear signal that the default setting was not effective for identifying the rare fraud cases. This is where the **Precision-Recall Curve** becomes the most important tool.

How I Arrived at the Score

The default 0.5 threshold was clearly not suitable. I needed to analyze the trade-offs at different thresholds to make a better business decision.

The table below compares the model's performance at three different example thresholds.

Confidence Threshold	Key Business Implication	Precision (approx.)	Recall (approx.)
0.80	Too Cautious: Misses almost all fraud cases to avoid false alarms.	Very High (>99%)	Very Low (<10%)
0.50 (Default)	Ineffective: Biased towards the majority class and fails to detect most fraud.	High	Very Low
0.15	Too Aggressive: Catches most fraud but creates too many false alarms, harming customers.	Low (<70%)	Very High (>95%)

After using the interactive threshold slider in Vertex AI, none of these results are ideal. My goal was to find a "sweet spot" that could maximize fraud detection while maintaining a high level of customer trust. So, I made the decision: **The selected confidence threshold is 0.75**. At this point, the model achieves an approximate **Precision of 96%** and a **Recall of 82%**.

This decision is justified because it strikes a good business balance. We can successfully stop 82% of fraudulent transactions, which greatly reduces financial loss. At the same time, a high precision of 96% means that when we do flag a transaction, we are very sure it is actually fraudulent. This minimizes the number of false positives and protects the customer experience.

Lessons Learned and Future Impact

This hands-on assignment highlighted several key lessons for future AI projects. First, effective cloud resource management is essential, as shown by initial quota issues, underscoring that MLOps involves not only algorithms but also infrastructure planning and cost management. Second, the choice of evaluation metric must align with the business problem, as standard metrics like accuracy can be misleading for imbalanced datasets, making metrics such as AUC PR more appropriate for tasks like fraud detection. Third, while AutoML tools like Vertex AI can rapidly generate high-quality models, humans still remain critical in setting appropriate optimization goals and interpreting results.

References

[1] Chaudhary, K., Yadav, J., & Mallick, B. (2012). [A review of Fraud Detection Techniques: Credit Card. International Journal of Computer Applications.](#)

[2] Krunić, V. (2020). [Succeeding with AI: How to make AI work for your business. Manning Publications.](#)

[3] Wang, H., Yang, J., Liang, G., Lee, Y., & Cao, Z. (2024). [Analyzing the Usability, Performance, and Cost-Efficiency of Deploying ML Models on BigQuery ML and Vertex AI in Google Cloud. Proceedings of the 8th International Conference on Cloud and Big Data Computing \(ICCBDC 2024\).](#)

[4] Bhadra, M. (2024). [Credit Card Fraud Detection \(Data set\). Kaggle.](#)